

After missing a three-peat, the Chiefs are planning to win another Super Bowl and are scheduling their games now. Several Chiefs fans have chosen to support them by relabeling their season calendars.

All the fans have chosen to relabel the days of the week with 7 Chiefs players who made the top 100 for 2024. Each fan used the same order of days, but chose their own date to start. For example, one fan made Mondays "Mahomesday", while another put "Mahomesday" on Fridays. The other days follow in the order shown.



Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Mahomesday	Jonesday	Kelceday	McDuffieday	Humphreyday	Thuneyday	Smithday

Input

You will receive three lines of text. The first will be a date in ISO yyyy-MM-dd format. The second will be the Chiefsday of the week for that date. The third will be a date in the future from the first date given.

```
2025-03-01
Humphreyday
2025-03-14
```

Output

Determine what Chiefsday of the week the date in the future will fall on, and also how many days there were between the starting date and the ending date (exclusive, do not count either of the dates given in the input for that range) and output that using the following format.

```
2025-03-14 is a McDuffieday
There are 12 days between the two dates
```

Discussion

There are a variety of ways to do this, including simply counting. Some programming languages have libraries which will make this somewhat trivial. Some don't. To help those of you using languages without comprehensive datetime libraries, we have supplied the days in each month of the U.S. calendar. Note that February can have 28 days, or 29 in leap years (in years that are multiples of 4).

Days in U.S. calendar months:

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
31	28/29	31	30	31	30	31	31	30	31	30	31

In the Additional Examples given below, the actual day of the week for 2024-12-25 is Wednesday, but it's labeled Kelceday, so when the real answer for 2025-01-31 would be Friday, replace it with Humphreyday. For 2028-03-05, the actual day of the week is Sunday, but Thuneyday is given. The actual day of the week of 2030-01-09 is Wednesday, so we output Jonesday. Important: Each dataset may use a different shift of Chiefsdays.

Reminder: have you run your solution against **all** of the student data sets?

Additional Examples

Input 2	Output 2
2024-12-25 Kelceday 2025-01-31	2025-01-31 is a Humphreyday There are 36 days between the two dates

Input 3	Output 3
2028-03-05 Thuneyday 2030-01-09	2030-01-09 is a Jonesday There are 674 days between the two dates